

Speed, Distance and Time — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Award method marks for a correct formula and a correct substitution even when the arithmetic fails. Almost every mark lost on this topic goes one of three ways: minutes treated as decimals of an hour, the m/s to km/h factor applied backwards, or two speeds averaged when the journey should have been. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 speed = distance ÷ time **[1]** [2]
SI unit: **m/s** (metres per second) **[1]**
- 2 (a) $100 \div 20 = 5 \text{ m/s}$ [4]
(b) $240 \div 3 = 80 \text{ km/h}$
(c) $45 \div 6$ **[1]** = **7.5 m/s [1]**
In (b) the answer is in km/h because the data was in km and hours. No conversion was asked for.
- 3 (a) $12 \times 30 = 360 \text{ m}$ [2]
(b) $60 \times 2 = 120 \text{ km}$
- 4 (a) $400 \div 8 = 50 \text{ s}$ [2]
(b) $150 \div 50 = 3 \text{ hours}$
- 5 (a) $20 \times 3.6 = 72 \text{ km/h}$ [3]
(b) $90 \div 3.6 = 25 \text{ m/s}$
(c) $5 \times 3.6 = 18 \text{ km/h}$
A quick check: 72 km/h is a believable road speed. Answers like 7200 or 5.6 signal the wrong direction.
- 6 (a) **0.5 h** [3]
(b) **2.25 h**
(c) **180 s**
(b) is the standard trap. 2 h 15 min is 2.25 h, never 2.15 h.
- 7 $45 \div 30$ **[1]** = **1.5 m/s [1]** [3]
It is constant because the distance increases by the same amount (15 m) in each equal 10 s interval **[1]**
- 8 The object is **stationary [1]** [2]
because its distance from the start is not changing as time passes **[1]**
"Constant speed" is the wrong answer here and is chosen more often than the right one.
- 9 15 minutes = 0.25 h **[1]** [3]
 $6 \div 0.25$ **[1]**
= **24 km/h [1]**
 $6 \div 15 = 0.4$ scores the method mark only. That answer is in km per minute.
- 10 50 m/s is about 180 km/h **[1]** [2]
That is far faster than any person can walk — a walking speed is closer to 1.5 m/s, so he has probably divided the wrong way round **[1]**
- 11 $15 \times 40 = 600 \text{ m [1]}$ [2]
= **0.6 km [1]**

- 12** Average speed is the total distance divided by the total time for a whole journey [1] [2]
Instantaneous speed is the speed at one particular moment, such as the reading on a speedometer [1]

Paper B — Medium

30 MARKS

- 1** (a) $80 \div 40$ [1] = **2 m/s** [1] [5]
(b) The runner is **stationary** — the distance stays at 80 m [1]
(c) She is **returning towards the start** [1]; 50 m in 20 s, so the speed is **2.5 m/s** [1]
In (c) she returns faster than she set out. Both the direction and the number are needed.
- 2** Axes labelled "time (s)" and "distance (m)" with sensible scales [1] [4]
Straight line from (0, 0) to (30, 300) [1]
Horizontal line from (30, 300) to (50, 300) [1]
Straight line from (50, 300) to (70, 400), visibly **less steep** than the first section [1]
The gradients are 10 m/s and 5 m/s, so the second sloped section must be noticeably shallower. A parallel final section loses that mark.
- 3** Three sentences, each containing a number [1] [4]
Section 1 described with distance or speed: 300 m in 30 s, at 10 m/s [1]
The stop described with its duration: stationary for 20 s [1]
Section 3 described: a further 100 m in 20 s, at 5 m/s [1]
- 4** Total distance = $20 + 30 = 50$ km [1] [4]
Total time = $0.5 + 1 = 1.5$ h [1]
 $50 \div 1.5$ [1]
= **33.3 km/h** [1]
- 5** Total distance = $15 + 10 = 25$ km [1] [4]
Total time = $30 + 15 + 15 = 60$ minutes = 1 h [1]
 $25 \div 1$ [1]
= **25 km/h** [1]
The rest must be included in the total time. Leaving it out gives 33.3 km/h.
- 6** The rest adds to the total time [1] [3]
but adds nothing to the total distance [1]
Average speed is total distance $\div$ total time, so a larger bottom with the same top gives a smaller answer [1]
- 7** $12 \times 3.6 = 43.2$ km/h [1] (or $40 \div 3.6 = 11.1$ m/s) [1] [3]
12 m/s is faster [1]
Either conversion direction is fine, provided both speeds end up in the same unit before they are compared.
- 8** The gradient represents the **speed** of the object [1] [3]
Pick two points where the line crosses gridlines exactly and read off their coordinates [1]
Divide the change in distance by the change in time, using the axis values rather than counting squares [1]

Paper C — Hard

30 MARKS

- 1** (a) $60 \div 60 = 1$ h [1]; $60 \div 120 = 0.5$ h [1] [5]
(b) Total 120 km in 1.5 h [1]; $120 \div 1.5 = 80$ km/h [1]
(c) The car spends twice as long at the slower speed, so 60 km/h counts for more of the journey than 120 km/h does [1]

- 2** (a) Distance = **1000 m [1]**; time = $50 + 200 = \mathbf{250\ s\ [1]}$ [5]
 (b) $1000 \div 250\ \mathbf{[1]} = \mathbf{4\ m/s\ [1]}$
 (c) He spends four times as long walking as running, so the slow speed dominates; the answer is much closer to 2.5 than to 10 **[1]**
Averaging the two speeds gives 6.25 m/s, more than half as big again as the truth.
- 3** (a) $330 \times 6\ \mathbf{[1]} = \mathbf{1980\ m\ [1]}$ [4]
 (b) **1.98 km [1]**
 (c) Light travels roughly a million times faster than sound, so it covers 1980 m in a tiny fraction of a second — far too short to notice **[1]**
- 4** $15\ \text{m/s} = 15 \times 3.6 = 54\ \text{km/h}\ \mathbf{[1]}$ [4]
 $840\ \text{m/min} = 840 \div 60 = 14\ \text{m/s}\ \mathbf{[1]}$, which is $14 \times 3.6 = 50.4\ \text{km/h}\ \mathbf{[1]}$
 Order: **50 km/h, 840 m/min, 15 m/s [1]**
The two slowest differ by only 0.4 km/h, so this cannot be settled by eye. Working is essential.
- 5** (a) Total time allowed = **4 h [1]**; time used = $100 \div 50 = \mathbf{2\ h\ [1]}$ [5]
 (b) 200 km still to travel, in 2 h remaining **[1]**
 (c) $200 \div 2\ \mathbf{[1]} = \mathbf{100\ km/h\ [1]}$
The first line of any target-average question should be total distance $\div$ target speed. Everything else follows from that time budget.
- 6** The two parts of the journey almost never take the same amount of time **[1]** [3]
 More time is spent travelling at the slower speed **[1]**
 so the slower speed counts for more, and the true average is always closer to it than the midpoint is **[1]**
- 7** The fastest section is the last one: 6 km between 15 and 20 minutes **[1]** [4]
 $5\ \text{minutes} = 1/12\ \text{h}\ \mathbf{[1]}$
 $6 \div (1/12)\ \mathbf{[1]}$
 $= \mathbf{72\ km/h\ [1]}$
Identifying the right section matters as much as the arithmetic. The first two sections are 3 km per 5 minutes; the third is a stop.

Paper D — Hard

30 MARKS

- 1** (a) $1000 \div 5 = \mathbf{200\ s\ [1]}$ [5]
 (b) Time used = $400 \div 4 = \mathbf{100\ s\ [1]}$; time remaining = $200 - 100 = \mathbf{100\ s\ [1]}$
 (c) 600 m in 100 s **[1] = 6 m/s [1]**
She must run the last 600 m half as fast again as the first 400 m, simply to lift her average by 1 m/s.
- 2** (a) 120 km in 2 h **[1] = 60 km/h [1]** [5]
 (b) Times are 1 h and 0.5 h, so 80 km in 1.5 h **[1] = 53.3 km/h [1]**
 (c) Driver A spends equal *times* at the two speeds, so the average is the midpoint. Driver B covers equal *distances*, which means longer spent at the slow speed, pulling the average below the midpoint **[1]**
This is the cleanest illustration in the topic of why "it depends what is being shared equally".
- 3** The car gains on the lorry at $25 - 20 = 5\ \text{m/s}\ \mathbf{[1]\ [1]}$ [4]
 $100 \div 5\ \mathbf{[1]}$
 $= \mathbf{20\ s\ [1]}$
Award both opening marks for any clear statement that only the difference in speeds matters.

- 4** The sound travels to the cliff and back, so it covers twice the distance **[1]** [4]
 Total distance = $340 \times 3 = 1020 \text{ m}$ **[1] [1]**
 Distance to the cliff = $1020 \div 2 = 510 \text{ m}$ **[1]**
Answering 1020 m scores three of four. The halving is the entire point of an echo question.
- 5** $72 \div 3.6 = 20 \text{ m/s}$ **[1] [1]** [4]
 20×90 **[1]**
= 1800 m [1]
Alternative route: $90 \text{ s} = 0.025 \text{ h}$, and $72 \times 0.025 = 1.8 \text{ km}$. Full marks either way.
- 6** (a) The straight line means a **constant speed [1]**; the curve getting steeper means the [4]
 object is **speeding up [1]**
 (b) Compare the steepness of the two lines at the end of the journey **[1]**; whichever is
 steeper there has the greater gradient and so the greater speed at that moment **[1]**
- 7** (i) The mistake: dividing by 20 without converting minutes into hours **[1]** [4]
 $20 \text{ min} = 1/3 \text{ h}$, so the speed is $30 \div (1/3) = 90 \text{ km/h}$ **[1]**
 (ii) The mistake: a flat line means the distance is not changing, so the car is
stationary, not moving steadily **[1]**
 A constant speed is shown by a straight line that **slopes [1]**